

104 學年度科學班甄選入學英文科 試題卷

一、詞彙 (10%)

說明：第 1 至 10 題，每題選出最適當的一個選項，標示在答案卡之「選擇題答案區」。
每題答對得 1 分，答錯不倒扣。

1. A _____ diet would help to reduce global warming, as the livestock industry has been blamed for creating a substantial percentage of the world's carbon dioxide emissions.
(A) balanced (B) low-fiber (C) vegetarian (D) flawless
2. As Sharon is shy and not sociable, she has great difficulty maintaining a(n) _____ relationship with others.
(A) intimate (B) honest (C) lukewarm (D) reliable
3. Many successful people are avid readers. Therefore, young people are often advised to read extensively not only to refresh their minds but also to broaden their _____.
(A) meanings (B) horizons (C) measures (D) resources
4. You have to work out how much you can _____ afford before making purchases so that you can live within means.
(A) initially (B) realistically (C) considerably (D) apparently
5. People often _____ their views because their ways of seeing things change as they grow older.
(A) convey (B) establish (C) release (D) modify
6. Regardless of all the obstacles and setbacks, Jeremy Lin, an American professional basketball player for the Los Angeles Lakers of the National Basketball Association (NBA), _____ followed his dream.
(A) hospitably (B) persistently (C) extremely (D) hesitantly
7. With the lowest birthrate in the world, Taiwan's population structure will _____ drastic changes in the next 30 years and become extremely aged.
(A) undergo (B) contain (C) resemble (D) expect
8. Teenagers who are _____ to smartphones spend much time online day in and day out, which may eventually affect their academic performance as well as their health.
(A) affectionate (B) allergic (C) alert (D) addicted
9. With the aid of the stars, the sailors managed to _____ across the stormy sea.
(A) negotiate (B) investigate (C) navigate (D) intimidate
10. The tradition of eating Yuan-Xiao on the Chinese Lantern Festival has its _____ in the

Eastern Jin Dynasty in the fourth century.

- (A) religion (B) collision (C) religion (D) origin

二、綜合測驗 (50%)

說明：第 11 至 35 題，每題一個空格，請依文意選出最適當的一個選項，標示在答案卡之「選擇題答案區」。每題答對得 2 分，答錯不倒扣。

11-15 為題組

(A) Spinach has long been known as the source of cartoon character Pop-eye's super strength, but recently a group of scientists believe that it also holds the __11__ clean, efficient and sustainable alternative fuel. Purdue University physicists, __12__ their international peers, are studying the proteins found in spinach, which convert sunlight into chemical energy.

"The proteins we study are part of the most efficient system ever built, __13__ converting the energy from the sun into chemical energy with an unrivaled 60 percent efficiency," says Yulia Pushkar, Assistant Professor of Physics at Purdue. Pushkar further states that __14__ how this system works is critical for creating artificial photosynthesis. Photosynthesis is the process in which plants __15__ sunlight to convert carbon dioxide and water into energy. If scientists could harness the power of photosynthesis, mankind would have access to an effective, efficient and renewable energy source.

11. (A) demand for (B) key to (C) talent in (D) position as
12. (A) regardless of (B) except for (C) in terms of (D) along with
13. (A) able to (B) capable of (C) depending on (D) resulting in
14. (A) understanding (B) understand (C) understood (D) to be understood
15. (A) shield (B) transmit (C) utilize (D) dissolve

16-20 為題組

(B) Paddling out to catch the new wave at a surfing contest, James had a heart attack, which knocked him off his surfboard. He inhaled water, lost __16__, and was soon floating face down in the ocean.

About the time, Hira Edmonds, a lifeguard and a senior instructor for a surf-safety company, was teaching 14 other lifeguards the finer points of emergency rescue. He spotted James drifting away from the main competition area. It didn't take long __17__ he realized someone was in peril. Edmonds rallied the team, grabbed first-aid equipment and rushed to

the water.

The lifeguards dragged James to the beach. More than a dozen rescuers __18__ pumping oxygen and performing chest compressions on the unresponsive surfer. For nearly an hour—far longer than a single lifeguard would have been able to continue the lifesaving technique—the team continued CPR. An ambulance finally arrived and took James to the hospital, __19__ he spent a week in intensive care while his brain and heart healed.

A month after his release, James reunited with six of the lifeguards who saved him. “I am much __20__ to all of them,” he said with a grateful smile. “Being spotted by a group of lifeguards was the only thing that saved me.”

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| 16. (A) consensus | (B) consciousness | (C) conscience | (D) censorship |
| 17. (A) while | (B) after | (C) until | (D) before |
| 18. (A) took turns | (B) had difficulty | (C) felt like | (D) came near |
| 19. (A) what | (B) which | (C) where | (D) there |
| 20. (A) adapted | (B) obliged | (C) attracted | (D) devoted |

21-25 為題組

(C) Exercise can do good not only to our physical well-being but also to our psychological health. In the first place, exercise can __21__ self-confidence. Setting up and meeting goals such as jogging thirty minutes a day or walking half a mile each day makes us feel stronger and gives us a sense of __22__. Moreover, physical activities cause the brain to release chemicals called endorphins, which creates a positive feeling, thereby __23__ our stress and anxiety. __24__, exercising gives us a good reason to go outdoors and interact with people. This helps dispel the sense of isolation that contributes __25__ depression. So, why not just get up from the couch and get some workout?

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| 21. (A) persuade | (B) reserve | (C) shrink | (D) enhance |
| 22. (A) achievement | (B) occasion | (C) humor | (D) direction |
| 23. (A) reduce | (B) reduces | (C) reducing | (D) reduced |
| 24. (A) For instance | (B) To sum up | (C) In fact | (D) Lastly |
| 25. (A) for | (B) to | (C) on | (D) at |

26-29 為題組

(D) At the age of 21, Stephen Hawking, one of the most famous theoretical physicists ever, was diagnosed with ALS, a form of Motor Neurone Disease. As time progresses, his physical abilities deteriorate __26__ this disease. In spite of the physical challenges, he never gives up on himself and his mental capacities accelerate over the years. Known for his work

__27__ black holes and for authoring several popular science books, Hawking has shed light on the formation of the cosmos, the nature of time, and the ultimate fate of the universe. In the Paralympics 2012 opening ceremony, with a speech-generating device __28__ to a single cheek muscle, Hawking encouraged the audience to "look up at the stars and not down at your feet" and to "be curious." Though he is almost entirely paralyzed and is __29__ to a wheelchair, this pre-eminent scientist is not disabled in spirit and his imagination and research has reached far into the outer space.

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| 26. (A) by means of | (B) on account of | (C) with access to | (D) in pursuit of |
| 27. (A) despite | (B) beyond | (C) regarding | (D) underneath |
| 28. (A) attach | (B) attaches | (C) attaching | (D) attached |
| 29. (A) confined | (B) contained | (C) confirmed | (D) consisted |

30-35 為題組

(E) All over the world, many people are enthralled by the stunning beauty of butterflies. __30__, not many people realize that more and more butterflies are becoming endangered as the wild places they inhabit are lost to development or as they fall victim to pesticides. In the U.S. alone, over twenty species of butterflies are on the brink of extinction, __31__ of the iconic and beloved monarch butterfly. Although that number may seem small, losing even one species would be tragic since butterflies play a critical role in __32__ the health of our environment. They __33__ help pollinate fruits, flowers and vegetables __33__ provide food for other animals. With a view to protecting the endangered butterflies from extinction, the American Museum of Natural History in New York holds a special butterfly exhibit __34__ the important role that butterflies play in our environment. Aside from the environmental groups, ordinary people can do their part to support and rescue these gentle __35__ simply by planting butterfly-friendly gardens. The world cannot afford to lose any butterflies. As environmentalist John Muir once commented, “When we try to pick out anything by itself, we find it hitched to everything else in the Universe.”

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| 30. (A) Nevertheless | (B) Moreover | (C) Hence | (D) Furthermore |
| 31. (A) include | (B) inclusive | (C) including | (D) included |
| 32. (A) draining | (B) remaining | (C) maintaining | (D) detaining |
| 33. (A) not merely; but also | (B) either ; or | (C) neither; nor | (D) never; but |
| 34. (A) highlight | (B) highlights | (C) highlighted | (D) highlighting |
| 35. (A) gestures | (B) treasures | (C) creatures | (D) measures |

三、文意選填 (30%)

說明：第 36 至 55 題，每題一個空格，請依文意在文章後所提供的(A)到(BC)選項中分別選出最適當者，並將其英文字母代號標示在答案卡之「選擇題答案區」(雙字母的選項請劃記兩個字母)。36 至 45 題每題答對得 1 分，46 至 55 題每題答對 2 分，答錯不倒扣。

36-45 為題組

(A) Luke Massella was 10 years old when he received his lab-grown bladder. He'd had health problems __36__ being born with spina bifida, a condition in which the spinal cord doesn't fully form. But at the age of nine, Luke's problems went from manageable to near __37__. His bladder pressure was getting really high, and it was pushing fluid back up into his kidneys; he was in kidney failure.

The new bladder saved his life. Rather than being transplanted from a donor or constructed from artificial parts, Luke's replacement organ had been grown in a lab using his own cells. Luckily for Luke, his urologist had known of a then-experimental technique __38__ by Professor Anthony Atala of the Wake Forest Institute for Regenerative Medicine in North Carolina, US.

Atala's team removed a small piece of Luke's bladder and then grew new cells on a scaffold. Two months later, the old bladder was removed and __39__ with the lab-grown organ. And it's been serving him well ever since, and he is now 23.

If we need a new hip or hand, we would normally have to build an artificial one. If we need a new heart or liver, we transplant the organ from a person who has __40__ died. But while organ transplants and prosthetic body parts have saved many lives, they're not a perfect solution. With transplants, for instance, there simply aren't enough to go round: in the UK, the __41__ waiting time for a heart transplant is about five years, for a lung it's over 14 years. And when if a patient is lucky enough to receive an organ from a donor, they face a lifetime on immunosuppressive drugs to prevent it being rejected as __42__ tissue.

To sidestep such complications, researchers are busy developing alternative approaches through regenerative medicine---techniques that __43__ to repair or regrow a patient's diseased or damaged body parts. While tissue engineers like Atala are able to grow almost every organ in the body, from livers, kidneys and hearts, other scientists are transplanting stem cells into damaged tissues and even reprogramming a patient's cells to __44__ their own organs. "Whether it involves building replacement organs or developing cell therapies to promote __45__, the ultimate goal is to improve function," said Atala.

(A) repair	(B) since	(C) aim	(D) recently	(E) fatal
(AB) healing	(AC) average	(AD) developed	(AE) replaced	(BC) foreign

46-55 為題組

(B) "I believe that a gun has no power at all because it can only kill, but a pen can give life. A pen can save lives." This quote from Malala Yousafzai, a Pakistani activist for female education, reveals how __46__ this young girl is. Malala was born in Mingora, __47__ in Pakistan's Swat Valley, in 1997 and attended school there. After the Taliban attacked girls' schools in Swat and deprived the girls __48__ their right to an education, Malala began blogging for the BBC about living under the Taliban's rule and about her urge to go back to school. When she was 14, Malala and her family learned that the Taliban had __49__ a death threat against her. Though Malala was frightened for the safety of her father, an anti-Taliban activist, she and her family __50__ felt that the fundamentalist group would not actually harm a child. Unexpectedly, on October 9, 2012, on her way home from school, a man boarded the bus Malala was riding in and __51__ to know which girl was Malala. When her friends looked toward Malala, her location was __52__. The gunman fired at her and the shooting left Malala in critical __53__. Fortunately, Malala survived the shocking assassination __54__, and has gone on to become an international role model. In 2014, Malala was announced as the co-recipient of the 2014 Noble Peace Prize for her struggle against the suppression of children and young people __55__ for her promotion of all children's right to education. At age 17, Malala Yousafzai is the youngest-ever Nobel Prize laureate.

(A) attempt	(B) issued	(C) inspirational	(D) as well as	(E) of
(AB) located	(AC) given away	(AD) condition	(AE) demanded	(BC) initially

四、閱讀測驗 (10%)

說明：第 56 至 60 題，每題請分別根據各篇文章之文意選出最適當的一個選項，標示在答案卡之「選擇題答案區」。每題答對得 2 分，答錯不倒扣。

56-58 為題組

(A) When you look in the mirror, do you like the way you look? Do you always find something you want smoothed, shaped, or improved? Obsession with body image is growing, but who on earth sets the standard for beauty for the public?

Ideas of beauty vary from culture to culture. However, due to the fast-spreading

modern media, some beauty standards have become extremely rigid, though sometimes unreal. Fashion magazines, TV commercials, and billboards bombard the public with thin-figured models. What is not emphasized, however, is that the few imperfections models may have are airbrushed away; stomachs are flatter, thighs are thinner, and all skin blemishes are removed. Airbrushed images gone wild create an unattainable beauty that might hurt people's self-esteem.

Recent photos posted online of model's before-and-after airbrushing have shocked viewers. Without Photoshop, they look pretty normal. Overly-airbrushed celebrity photos place the issue of digitally-enhanced images in the spotlight. Unfortunately, the stakes are much bigger than celebrity gossip.

A report, "The Impact of Media Images on Body Image and Behaviors," revealed that on average, 3,000 images a day feature heavily-airbrushed models whose beauty is not real. Young girls are affected by the twisted images the worst. These images can cause problems such as eating disorders, depression and low self-esteem.

It's not only female adults and adolescents that suffer the negative effects. Children as young as grade school age often want thinner bodies after exposure to Barbie dolls and teen magazines. Even men are falling victim to it. The result is higher sales in cosmetics and dieting products, but a drop in self-confidence. The long-term impact isn't pretty.

Noticing the problem, in 2009 a U.K. political party started *the Real Woman campaign* against advertising companies. They demanded more honest advertisements and placing labels on airbrushed images. The label informs people the image has been changed and not real. Last, airbrushed ads targeting children under 16 are no longer allowed.

Beauty is only skin-deep. Having a positive body image and confidence is very important to one's happiness, so embrace who you are. After all, beauty is in the eye of the beholder.

56. According to the passage, which of the following might influence our standards of beauty?

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| (A) Cosmetics manufacturers | (B) School authorities |
| (C) Cultures and values | (D) Fashion magazines |

57. Which of the following is **not** listed as one of the negative effects digitally-enhanced images have on the public?

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|-----------------------------|---------------------|
| (A) A drop in make-up sales | (B) Low self-esteem |
| (C) Eating disorders | (D) Mental diseases |

58. What did *the Real Woman campaign* aim to do?

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|---|------------------------------------|
| (A) To reinforce the virtue of honesty. | (B) To promote sales in cosmetics. |
| (C) To boost women's confidence. | (D) To ask for more honest photos. |

59-60 為題組

(B) We live in a watery world, but climate change and population growth are leading the planet toward a water crisis. Humans will have to find smart ways to avoid wasting water.

About 70-73% of earth is covered by water so it is easy to believe that water is abundant and never ending. However, **looks can be deceiving**. Only one percent of all of the water on Earth is fresh, liquid, and usable! Most of the world's H₂O is salty ocean water or frozen polar ice.

Everyone needs safe drinking water, but about 1.1 billion people have no access to clean water. By 2050, the world's population could grow from 6.7 billion to more than 9 billion. All of those people will need water to drink, to grow crops and to produce energy.

Another noticeable fact is climate change. Because of climate change, some areas of the world are growing drier. The total area of the Earth's surface classified as very dry has doubled since the 1970s. The combination of climate change and increased demand is pushing the world toward a crisis over water. "You do the math, and it gets a little scary," says Stuart Minchin, a water expert in Australia.

Water is wasted in both rich and poor countries. The precious resource is misused in industry, in farming and in homes. Peter Gleick, head of the Pacific Institute, an environmental group in Oakland, California, says it's time for bold steps to use less water. "We're waking up," he says. "But not fast enough."

59. The article is mainly about _____.

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| (A) how climate change is leading to the expansion of the desert area |
| (B) what people around the world are doing to deal with water shortage |
| (C) how climate change and population growth are bringing about a water crisis |
| (D) why we should control the world's birth rate and stop the population growth |

60. The words "looks can be deceiving" in the second paragraph imply that _____.

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| (A) in fact a large part of the surface of the earth is not covered by water |
| (B) water resources are not that plentiful as we might have assumed |
| (C) the earth is actually not blue as photographed from the outer space |
| (D) it is not hard to perceive that we can desalinate water |

試題結束